

PRD: Frontend Web Application — Cloud Storage / Sharing / Streaming Platform

Status: Draft for handoff to design/build tool

Stack: React + Vite

Format: Full web application (not a stripped-down mobile view) with highly refined, mobile-responsive design — must work excellently from large desktop screens down to small mobile viewports within a single responsive codebase.

Companion document: See `PRD-02-Backend-Platform.md` for the API/data model this frontend integrates with. This document focuses on user-facing screens, flows, and UX requirements — not backend implementation.

1. Problem Statement

Users need a polished, trustworthy web interface for uploading, organizing, sharing, and streaming files — one that feels as refined as Google Drive/Dropbox despite being a leaner, storage-generous, ad-

supported product. The interface must work equally well as a primary desktop workspace and as a fully-featured mobile browsing experience, since no native mobile app exists yet.

2. Goals

- Deliver a **single responsive web app** (not separate mobile/desktop builds) that adapts cleanly across breakpoints — desktop, tablet, and mobile — with mobile treated as a first-class experience, not an afterthought.
- Make the core loop (upload → organize → share → stream/preview) fast and low-friction, since this is what drives activation.
- Present ads (banner, interstitial, video pre-roll) in a way that funds the free tier without feeling intrusive or degrading trust — ads should feel like a fair exchange, not a dark pattern.
- Make paid-tier value (more storage, ad-free, HD streaming, password-protected links) visible and desirable at the right moments, without being pushy.

3. Non-Goals

- Native mobile app UI (iOS/Android) — this is a web-only build in this phase; a highly responsive

web app is the mobile experience for now.

- Desktop sync client UI — backlog, not part of this build.
- Creator monetization / earnings dashboard — deferred feature, not designed in this phase.
- Multi-language UI — English only for this phase.
- Admin panel UI — internal moderation/admin tooling is handled via Django admin (see backend PRD), not a custom frontend.

4. Target Users & Key Personas

- **Everyday storage user** — backs up photos/documents, organizes into folders, occasionally shares with friends/family. Primary driver of free-tier volume.
- **Sharer/broadcaster** — creates channels, shares content to many people at once (not primarily seeking discovery/boost, per product research — channels are used as a broadcast tool).
- **Heavy consumer** — browses discoverable content, streams video, downloads shared files; likely to encounter the most ad inventory.
- **Paid upgrader** — has hit quota limits, wants ad-free/HD experience, or values password-protected sharing.

5. Information Architecture / Core Screens

1. **Auth flow** — Sign up (phone+OTP primary, email/password and Google/Apple as alternatives), DOB capture (age-gating, not exposed as a "why do you need this" friction point — frame simply), recovery email setup (encouraged, not blocking).
2. **Dashboard / My Storage** — Root-level file/folder browser, quota usage indicator (visually clear: used vs. total, tier-aware), upload entry point, dismissible banner ad placement.
3. **Folder view** — Nested folder navigation (breadcrumb-based), grid/list toggle, per-item actions (share, rename, move, delete, download, preview).
4. **Upload flow** — Drag-and-drop + file picker, progress indication (supports large/resumable uploads), upload-conflict resolution UI ("Replace" vs "Keep both"), interstitial ad placement on upload action (free tier).
5. **File preview / Document viewer** — Unified preview surface handling: text/code (`.txt` / `.md` / `.json` / `.yaml`), images, PDF (PDF.js), and Office docs (`.doc` / `.ppt` rendered as an image-carousel of pages/slides — not a PDF, note this affects viewer pagination UI).

6. **Video player** — Custom-built player UI (not a third-party embedded widget look) consuming an HLS stream; resolution indicator/selector (capped for free tier); pre-roll ad sequencing before playback starts (frequency: 1 ad for 5–10 min videos, +1 per additional 15 min; first ad unskippable ~30s, later ones skippable after 5–10s).
7. **Sharing UI** — Generate private share (specific people) or public link; toggle discoverable; set optional link expiry; password-protect link (paid-tier gated, with a clear upsell moment if attempted on free tier); mature-content self-tagging toggle when making content discoverable.
8. **Discovery / Search** — Browse/search discoverable files, folders, and public channels; mature content shown blurred with an explicit age-confirmation gate (18+ only, hidden entirely for accounts under 18 and — per platform-specific policy — hidden from web sessions flagged as iOS clients, though this mainly affects native iOS which doesn't exist yet; still worth building the filtering hook now).
9. **Channels** — Channel creation, subscribe/unsubscribe, channel feed (posts = shared files/folders with captions), member/role management for owners (promote subscriber to

admin/posting rights), public vs. private visibility setting at creation.

10. **Trash** — List of soft-deleted items, restore action, retention countdown indicator (7 days free / 30 days paid — make the tier difference visible as a soft upsell).
11. **Notifications** — In-app notification center (bell icon + list), covering channel activity, share activity, quota warnings, security alerts.
12. **Account settings** — Profile (phone, recovery email, DOB), storage region selection, device backup toggle (+ Wi-Fi-only sub-toggle), active devices/sessions list, referral code/link + bonus storage tracker (with expiry countdown per TeraBox-style 180-day grants), subscription management (plan, billing history/invoices, cancel/upgrade), data export request, account deletion request.
13. **Upgrade / Pricing screen** — Clear tier comparison (Free 500GB vs. ₹99/2TB vs. ₹249/5TB), highlighting ad-free + HD streaming + password links as paid differentiators — not just storage size.
14. **Content reporting** — Lightweight report flow accessible from any discoverable file/folder/channel (reason selection, submit) — no user-blocking UI needed (product decision: users don't see other users directly).

6. Requirements

P0 — Must-Have

- [] Fully responsive layout — one codebase, breakpoints covering desktop / tablet / mobile, with mobile treated as a primary use case (not a cut-down view)
- [] All 14 screens/flows in Section 5, wired to the backend API described in the companion PRD
- [] Ad placements integrated per spec (banner, interstitial, pre-roll) — free tier only, cleanly hidden for paid users
- [] Quota/tier visibility woven throughout (dashboard, trash, upgrade prompts) without being nagging
- [] Age-gating UX for mature/discoverable content (blur + confirm, not a jarring block)
- [] Upload flow supports large files with resumable/progress UI
- [] Document viewer handles all specified formats correctly, including the Office-doc-as-image-carousel pattern (this is a deliberate product decision, not a fallback — build it as a first-class pattern, not an edge case)
- [] Custom video player (not an embedded third-party widget) — full control over player chrome/branding

P1 — Nice-to-Have

- [] Referral sharing UI polish (easy copy/share of referral link, visual progress toward 1TB cap)
- [] Notification preferences (which types trigger push vs. email, within the backend's defined categories)

P2 — Future Considerations

- [] Native app-like PWA enhancements (offline access, install prompt) — not required now, but avoid architectural choices that would block this later
- [] Creator earnings dashboard (when that program ships)

7. Design Direction Notes

- This is **not** a mobile app shrunk into a browser — it should read as a proper, complete web application on desktop, that also happens to be excellent on mobile. Avoid overly app-like chrome (bottom tab bars mimicking native apps) unless it genuinely serves the mobile experience better than a responsive adaptation of the desktop layout would.
- Ad placements need enough visual separation from core UI that they don't feel like

phishing/malware-adjacent patterns — a known trust risk for storage apps carrying ads (unlike ad-free competitors like Google Drive).

- The paid-tier upsell moments (quota nearing limit, attempting password-protected link on free tier, seeing an ad) should feel like natural, low-pressure nudges, not blocking interruptions — this is a trust-building product in a category where trust is the biggest competitive gap against Google Drive/Dropbox.

8. Open Questions

- **[Design]** Exact visual/brand identity — no brand name/domain finalized yet (per product owner, to be provided later); build against a placeholder brand system that's easy to re-skin.
- **[Product]** Whether the web app should visually differentiate itself when the user is likely on a mobile device in ways beyond responsive layout (e.g., app-install prompts, "add to home screen" nudges) — not decided.

9. Success Metrics

- Task completion rate for core flows (upload → share, upload → discoverable) in usability testing
- Time-to-first-upload for new signups

- Ad viewability/engagement without measurable drop in session length or activation (watch for ad-fatigue signals)
- Free-to-paid conversion rate attributable to in-context upsell moments (quota warnings, password-link attempts)